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COURSE CODE: SBT-203-BASIC NETWORKING SKILLS
LAB NAME: LAB1 HTTP ANALYSIS USING WIRESHARK
DATE: 08/09/26

SCENARIO

Your practical context

Act as a junior network forensic analyst and document a controlled plaintext HTTP session. Create a local training webpage, capture the traffic, reconstruct the TCP conversation, identify the HTTP request and response, and explain the evidential value of HTTP, TCP, IP and link-layer metadata.

OBJECTIVES

What you must demonstrate

Explain HTTP/TCP/IP forensic artefacts; capture a complete HTTP session; identify SYN, SYN-ACK and ACK; extract ports, sequence and acknowledgement numbers, timestamps and HTTP headers; reconstruct the session; document integrity hashes and a concise forensic timeline.

INSTRUCTOR BRIEFING

Laboratory guidance

SBT-DF203 — Basic Networking Skills for Digital Forensics

Lab 1 — HTTP Analysis Using Wireshark: Text Traffic

Official Lab Manual

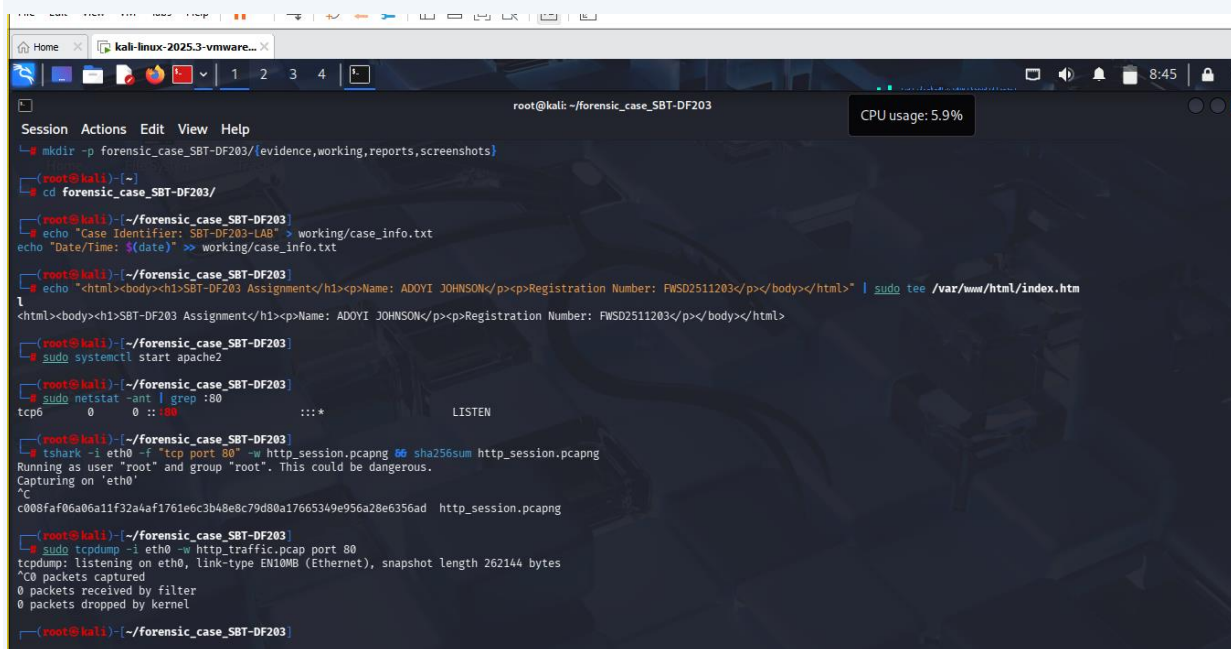
Read the complete manual before starting. [Open the official SBT-DF203 Lab 1 manual.](#)

The manual contains the detailed commands, evidence worksheets, marking rubric, screenshot checklist, safety controls and report structure. The portal instructions below are the assessed workflow summary and do not replace the manual.

Required Practical Workflow

1. Read the complete official lab manual before running any command.
2. Create the required evidence, working, reports and screenshots folders and record the case/lab identifier.
3. Create the authorised local Apache webpage containing your name and registration number and verify TCP port 80.
4. Capture the HTTP session with Wireshark or TShark, preserve the original PCAP/PCAPNG and calculate SHA-256 hashes.
5. Analyse the TCP three-way handshake, HTTP GET request, HTTP response, encapsulation and connection termination.
6. Prepare the required findings table, screenshots and professional report.

ANSWER:



```
root@kali: ~/forensic_case_SBT-DF203
CPU usage: 5.9%

Session Actions Edit View Help
root@kali:~/forensic_case_SBT-DF203
└─$ mkdir -p forensic_case_SBT-DF203/{evidence,working,reports,screenshots}

root@kali:~/forensic_case_SBT-DF203
└─$ cd forensic_case_SBT-DF203/

root@kali:~/forensic_case_SBT-DF203
└─$ echo "Case Identifier: SBT-DF203-LAB" > working/case_info.txt
└─$ echo "Date/Time: $(date)" >> working/case_info.txt

root@kali:~/forensic_case_SBT-DF203
└─$ echo "<html><body><h1>SBT-DF203 Assignment</h1><p>Name: ADOYI JOHNSON</p><p>Registration Number: FWSD2511203</p></body></html>" | sudo tee /var/www/html/index.htm

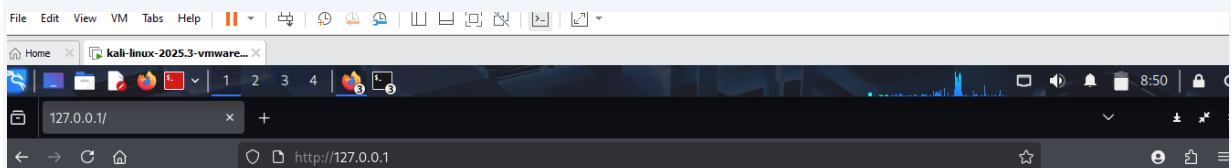
root@kali:~/forensic_case_SBT-DF203
└─$ sudo systemctl start apache2

root@kali:~/forensic_case_SBT-DF203
└─$ sudo netstat -ant | grep :80
tcp6      0      0  :::*          :::*          LISTEN

root@kali:~/forensic_case_SBT-DF203
└─$ tshark -i eth0 -f "tcp port 80" -w http_session.pcapng && sha256sum http_session.pcapng
Running as user "root" and group "root". This could be dangerous.
Capturing on 'eth0'
^C
c008faf06a06a1f32a4af1761e6c3b48e8c79d80a17665349e956a28e6356ad http_session.pcapng

root@kali:~/forensic_case_SBT-DF203
└─$ sudo tcpdump -i eth0 -w http_traffic.pcap port 80
tcpdump: listening on eth0, link-type EN10MB (Ethernet), snapshot length 262144 bytes
^C0 packets captured
0 packets received by filter
0 packets dropped by kernel

root@kali:~/forensic_case_SBT-DF203
```



SBT-DF203 Assignment

Name: ADOYI JOHNSON

Registration Number: FWSD2511203

Authorised Training Resource

Optional basic HTTP training capture

Use this resource only for the SBT-DF203 authorised training exercise. Preserve the source file and work from a verified copy where the manual requires it.

Evidence and Submission Package

- One PDF forensic report named SBT-DF203-Lab1_RegNo_FullName.pdf.
- Original/generated PCAP or PCAPNG evidence plus SHA-256 hash record.
- Reports/command-output files and required screenshots.
- One ZIP package containing the report and evidence outputs.